

Embedded systems exp 17

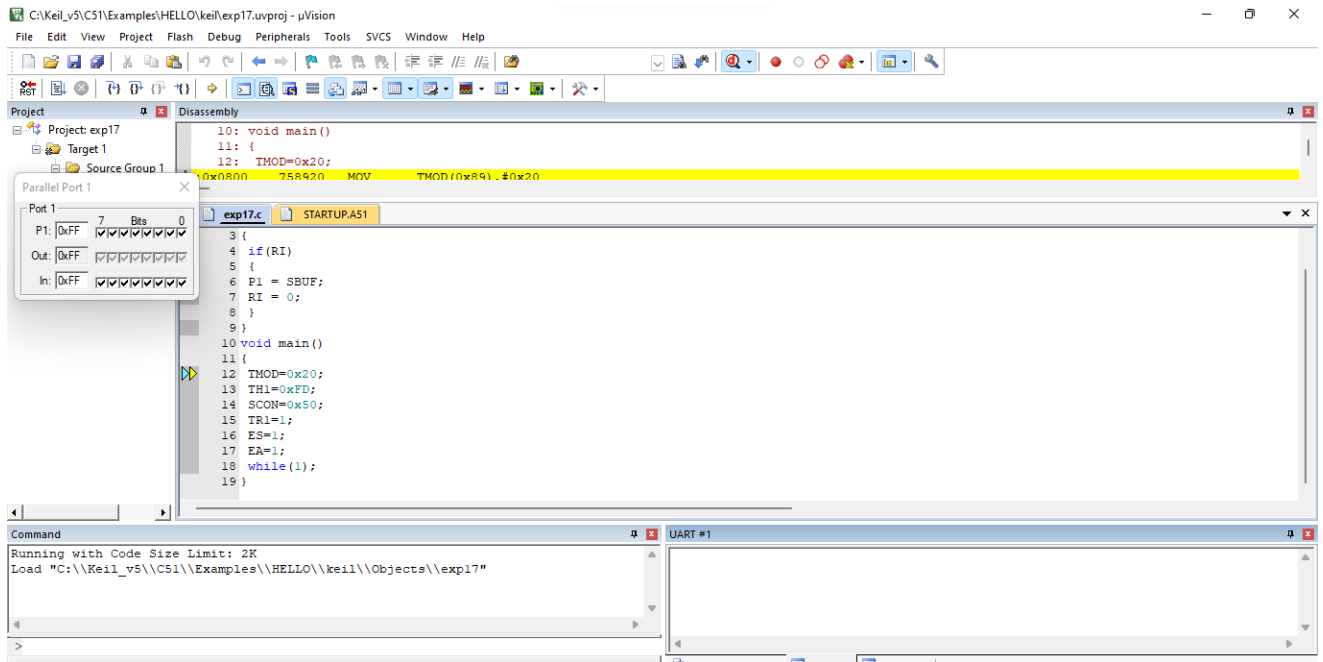
Program:

```
#include <reg51.h>

void serial_ISR(void) interrupt 4
{
    if(RI)
    {
        P1 = SBUF;
        RI = 0;
    }
}

void main()
{
    TMOD=0x20;
    TH1=0xFD;
    SCON=0x50;
    TR1=1;
    ES=1;
    EA=1;
    while(1);
}
```

Before interrupt



After interrupt

C:\Keil_v5\CS1\Examples\HELLO\keil\exp17.uvproj - µVision

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Registers

Register	Value
r0	0x00

Parallel Port 1

Port 1

P1: 0x41 7 Bits 0

Out: 0x41

In: 0xFF

Disassembly

```
18: while(1);
133: 0x080F 80FE SJMP C:080F
133: 0x0811 787F MOV R0,#IDATALEN - 1
133: 0x0811 787F MOV R0,#0x7F
```

exp17.c STARTUP.A51

```
3 {
4 if(RI)
5 {
6 P1 = SBUF;
7 RI = 0;
8 }
9 }
10 void main()
11 {
12 TMOD=0x20;
13 TH1=0xFD;
14 SCON=0x50;
15 TR1=1;
16 ES=1;
17 EA=1;
18 while(1);
19 }
```

Command

Running with Code Size Limit: 2K

Load "C:\Keil_v5\CS1\Examples\HELLO\keil\Objects\exp17"

UART #1